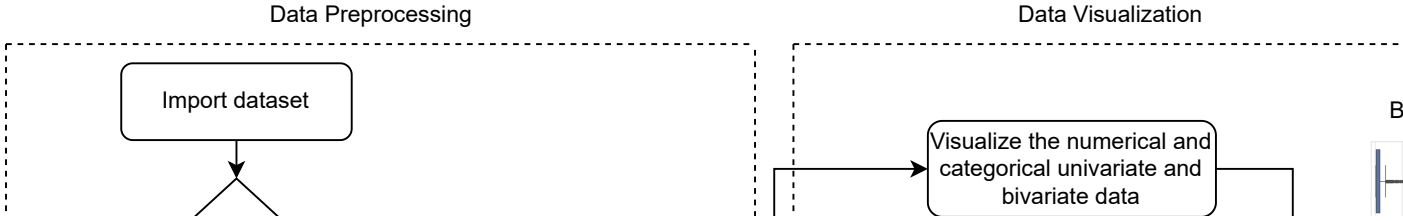
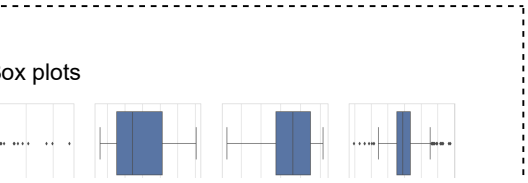
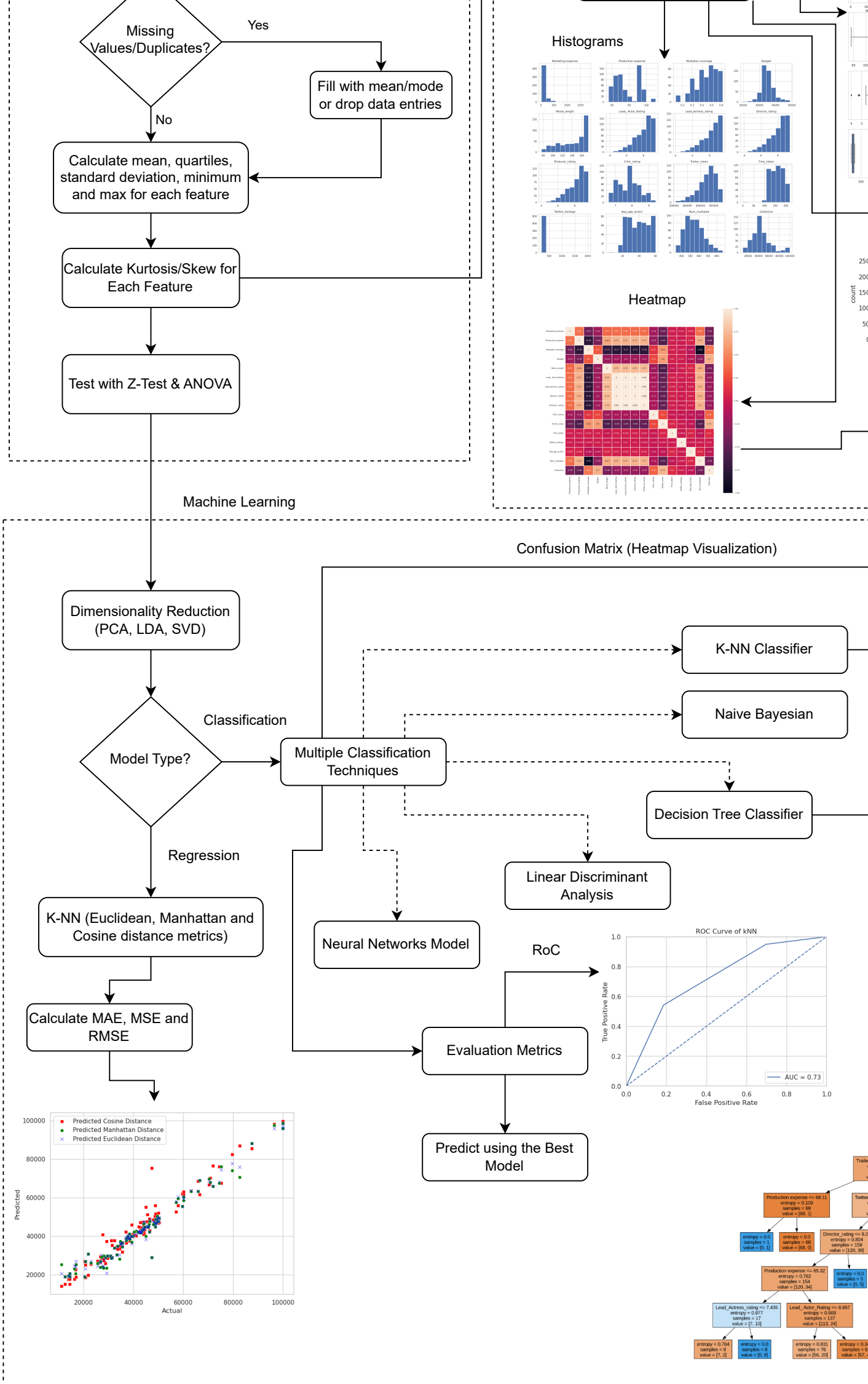
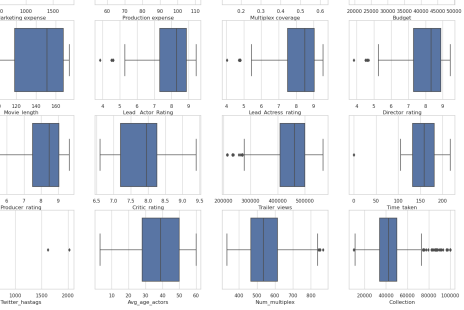
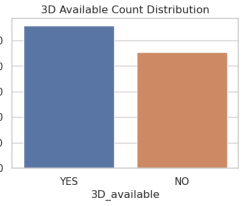

Data Preprocessing
Data Visualization
Import dataset
Visualize the numerical and categorical univariate and bivariate data

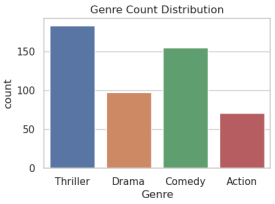

ox plots

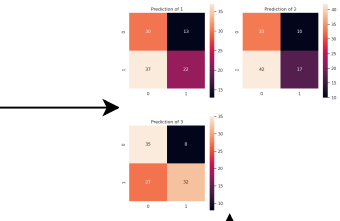

Missing Values/Duplicates?
Yes
No
Fill with mean/mode or drop data entries
Calculate mean, quartiles, standard deviation, minimum and max for each feature
Calculate Kurtosis/Skew for Each Feature
Test with Z-Test & ANOVA
Histograms
Heatmap
Machine Learning
Dimensionality Reduction (PCA, LDA, SVD)
Model Type?
Classification
Regression
Multiple Classification Techniques
Confusion Matrix (Heatmap Visualization)
K-NN Classifier
Naive Bayesian
Decision Tree Classifier
Linear Discriminant Analysis
Neural Networks Model
K-NN (Euclidean, Manhattan and Cosine distance metrics)
Calculate MAE, MSE and RMSE
Evaluation Metrics
RoC
ROC Curve of kNN
True Positive Rate
False Positive Rate
AUC = 0.73
Predict using the Best Model
Predicted Cosine Distance
Predicted Manhattan Distance
Predicted Euclidean Distance
Predicted
Actual



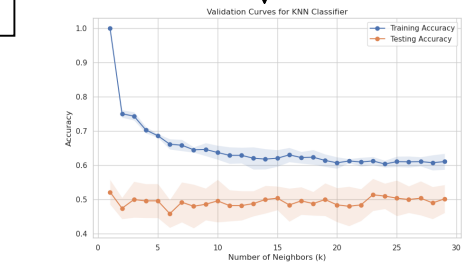
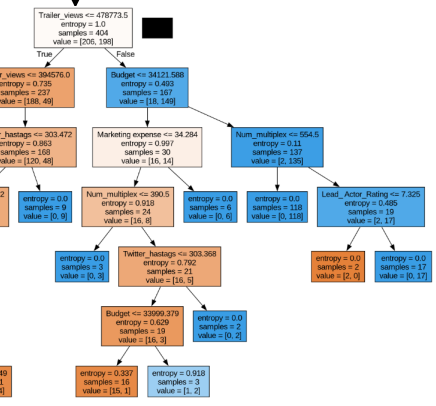
Bar plots







Estimating Best K



Decision Tree (Max Depth 7)